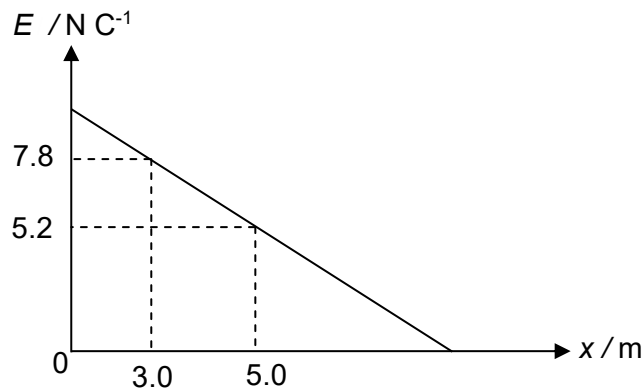


- 19** The graph below shows how the electric field strength  $E$  varies with displacement  $x$  from a point A. What is the change in potential for an electron if it is moved from a point 3.0 m away from A to another point 5.0 m away from A?



**A** 13 V

**B** - 13 V

**C** 1.3 V

**D** - 1.3 V

**Ans: B**

Electric Field Strength,  $E = -\frac{dV}{dr}$

$\Delta V = -\int E dr$  Hence *magnitude* of the change in potential is the area under this graph. Because  $E$  is positive here, the change in potential must be negative (a decrease)

$$dv = \frac{1}{2}(7.8 + 5.2)(5.0 - 3.0) = 13 \text{ V (a decrease in the positive } x \text{ direction)}$$